

Country Development Clustering and Budget Allocation Analysis

Comprehensive Report

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Executive Summary

This report presents a comprehensive analysis of **167 countries** based on **9 development indicators**, combining Exploratory Data Analysis (EDA) with K-Means clustering to identify **5 socioeconomic development profiles** and inform an illustrative **\$1 billion budget allocation strategy**.

Key Findings

- Data Quality:** Complete dataset with 0% missing values; Yeo-Johnson transformation reduced average skewness from 2.23 to 0.08 (96.4% improvement)
- Optimal Clusters:** 5 profiles selected for interpretability and policy relevance (silhouette score: 0.2226, PCA variance: 71.6%)
- Strongest Differentiators:** Child mortality ($\eta^2 = 0.848$), GDP per capita ($\eta^2 = 0.778$), Income ($\eta^2 = 0.769$)
- Largest Vulnerability Profile:** 44 countries (26.3%) exhibit the strongest combination of adverse health, demographic, and economic indicators
- Strongest Overall Profile:** 31 countries (18.6%) with high development outcomes
- Budget Allocation:** Need-based distribution from 2% to 45% per cluster, prioritizing the most vulnerable

Quick Reference

Metric	Result
Countries Analyzed	167
Development Indicators	9
Number of Clusters	5
Most Vulnerable	44 countries (26.3%)
Most Advanced	31 countries (18.6%)
Illustrative Budget	\$1,000,000,000
Largest Allocation	45% (\$450M) to High Development Vulnerability

The analysis provides a data-driven foundation for policy discussion, resource allocation planning, and stakeholder engagement.

1. Introduction

1.1 Project Overview

This project analyzes country-level development indicators to:

- Explore** the distribution, correlations, and outliers of 9 development indicators across 167 countries
- Transform** skewed features using Yeo-Johnson transformation for improved statistical properties
- Identify meaningful data-driven groupings** of countries based on their development patterns using K-Means clustering
- Characterize** each development cluster by its key socioeconomic features
- Allocate an illustrative \$1 billion development budget** based on cluster needs and policy priorities

1.2 Dataset Information

Aspect	Detail
Source	Country development indicators dataset
Countries	167 countries
Features	9 development indicators
Methods	EDA, Yeo-Johnson transformation, K-Means clustering, ANOVA
Goal	Cluster countries and inform an illustrative \$1 billion development budget

1.3 Features Analyzed

Feature	Description	Type	Range (Min-Max)
child_mort	Child mortality rate (per 1,000 births)	Health	2.60 – 208.00
exports	Exports as % of GDP	Economic	0.11 – 200.00
health	Health expenditure per capita	Health	1.81 – 17.90
imports	Imports as % of GDP	Economic	0.07 – 174.00
income	Net income per person (USD)	Economic	609 – 125,000
inflation	Annual inflation rate (%)	Economic	-4.21 – 104.00
life_expec	Life expectancy at birth (years)	Health	32.10 – 82.80
total_fer	Total fertility rate (births per woman)	Social	1.15 – 7.49
gdpp	GDP per capita (USD)	Economic	231 – 105,000

1.4 Report Structure

This report is organized as follows:

Section	Content
Sections 2–6	Exploratory Data Analysis (data quality, distributions, correlations, outliers, transformation, preprocessing)
Section 7	Key Findings Summary from EDA
Section 8	Clustering Analysis (optimal k selection, final cluster distribution, PCA visualization, cluster profiles, statistical validation)
Section 9	Geographic Distribution of clusters
Section 10	Budget Allocation Strategy (allocation framework, summary, policy area breakdown, Sankey diagram, caveats)
Section 11	Policy Recommendations (cluster-specific interventions, cross-cutting recommendations)
Section 12	Limitations and Caveats (methodological caveats, future work)
Section 13	Key Findings & Insights (major findings, final results at a glance)
Section 14	Conclusion
Section 15	Appendix (color legend, transformation details, ANOVA results, country lists)

2. Data Quality Assessment

2.1 Missing Values Analysis

The dataset demonstrated **exceptional data quality** with no missing values across all 167 observations and 9 features.

Quality Check	Result
Missing Values	✅ None found (0%)
Duplicate Rows	✅ 0 duplicates
Complete Rows	✅ 167 (100.0%)

2.2 Statistical Summary

The descriptive statistics reveal substantial variation across development indicators.

Descriptive Statistics

Feature	Mean	Std	Min	Q1	Median	Q3	Max	Skewness
child_mort	38.27	40.33	2.60	8.25	19.30	62.10	208.00	1.45
exports	41.11	27.41	0.11	23.80	35.00	51.35	200.00	2.45
health	6.82	2.75	1.81	4.92	6.32	8.60	17.90	0.71
imports	46.89	24.21	0.07	30.20	43.30	58.75	174.00	1.91
income	17,144.69	19,278.07	609	3,355	9,960	22,800	125,000	2.23
inflation	7.78	10.57	-4.21	1.81	5.39	10.75	104.00	5.15
life_expec	70.56	8.89	32.10	65.30	73.10	76.80	82.80	-0.97
total_fer	2.95	1.51	1.15	1.80	2.41	3.88	7.49	0.97
gdpp	12,964.16	18,328.70	231	1,330	4,660	14,050	105,000	2.22

Key Observations

- High Inequality:** Income and GDP per capita show extreme ranges (609 to 125,000 USD)
- Health Disparities:** Child mortality ranges from 2.6 to 208 per 1,000 births
- Economic Variation:** Inflation ranges from -4.21% to 104% (Nigeria)
- Skewed Distributions:** Six features exhibit significant positive skewness (> 1.0)

3. Exploratory Data Analysis

3.1 Feature Distributions

The distributions of key development indicators reveal substantial variation across countries.

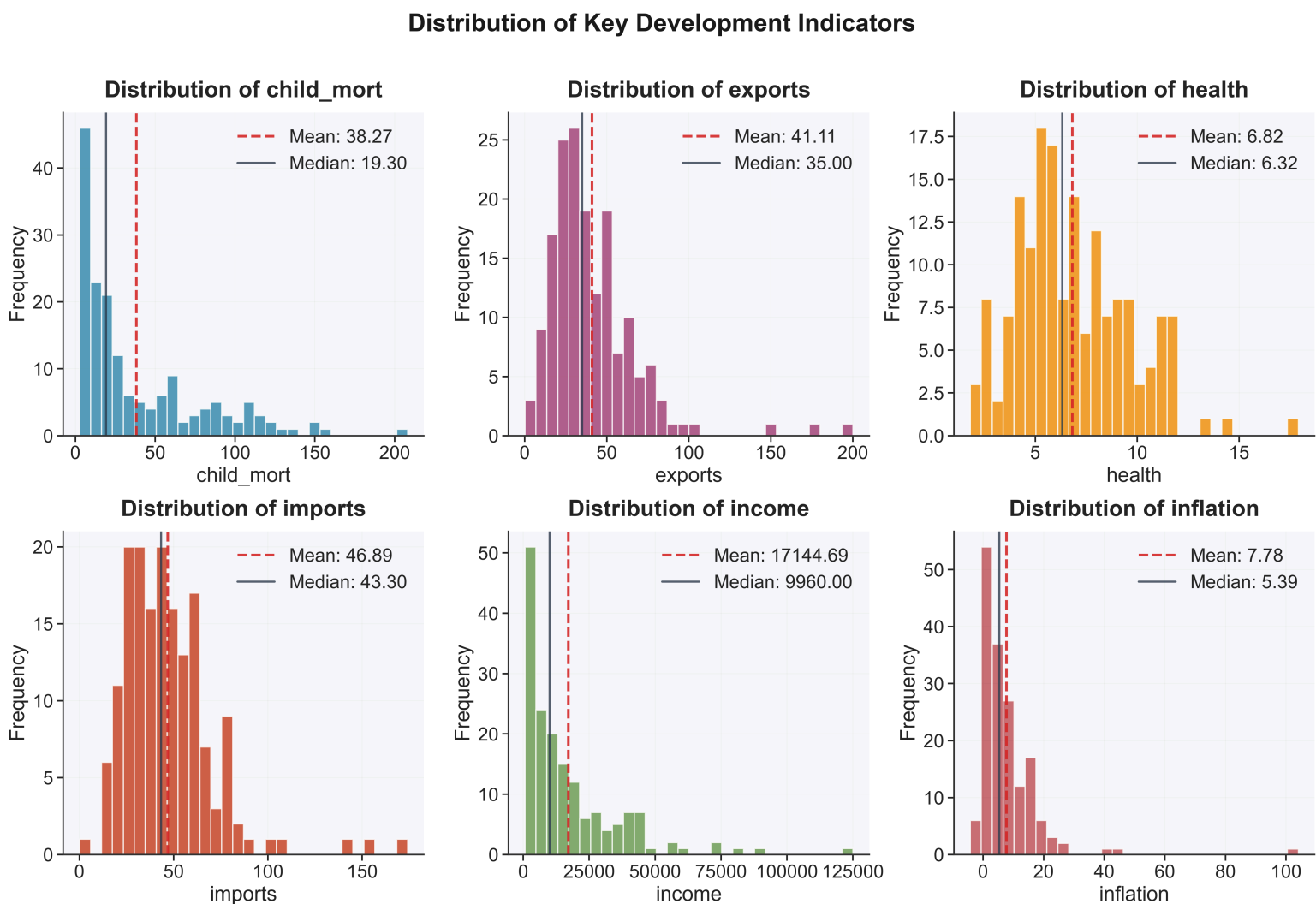


Figure 1: Distribution of key development indicators across 167 countries

Distribution Insights

- **Child Mortality:** Right-skewed, with most countries below 50 per 1,000, but a long tail of high-mortality countries
- **Income & GDP:** Highly skewed, with a few high-income countries (Luxembourg, Qatar, Singapore)
- **Inflation:** Most countries below 10%, with extreme outliers (Nigeria at 104%)
- **Life Expectancy:** Approximately normal, with a left tail of countries with low life expectancy
- **Total Fertility:** Range from 1.15 to 7.49, showing demographic diversity

3.2 Correlation Analysis

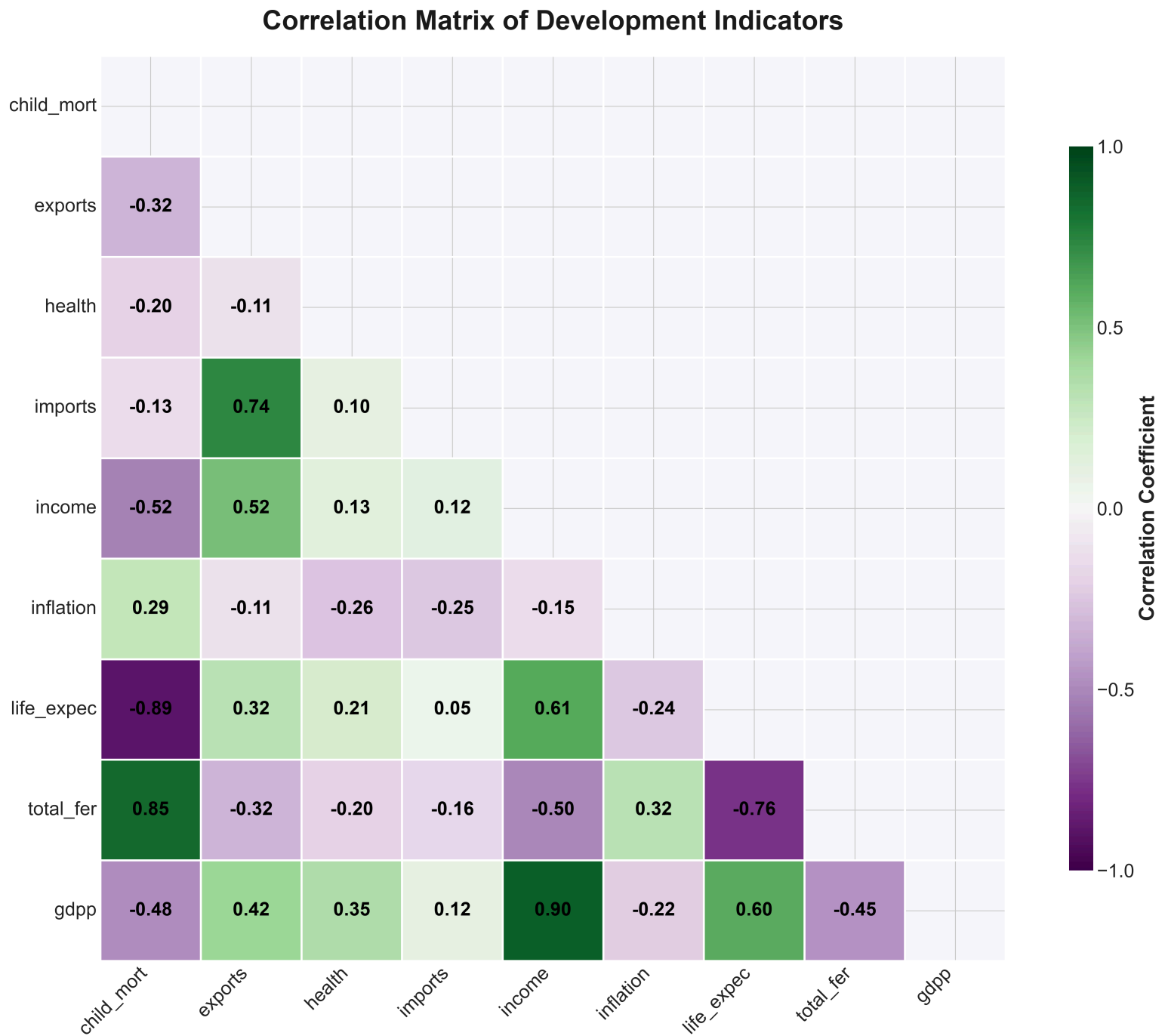


Figure 2: Correlation matrix of development indicators

Strong Correlations ($|r| > 0.8$)

Feature Pair	Correlation	Interpretation
child_mort ↔ life_expec	-0.887	Strong negative relationship
child_mort ↔ total_fer	+0.848	Strong positive relationship
income ↔ gdpp	+0.896	Strong positive relationship

Additional Notable Correlations

Feature Pair	Correlation	Interpretation
income ↔ life_expec	+0.799	Strong positive relationship
gdpp ↔ life_expec	+0.729	Strong positive relationship
child_mort ↔ income	-0.703	Strong negative relationship
total_fer ↔ life_expec	-0.739	Strong negative relationship

Correlation Insights

- Health-Economic Link:** Countries with higher income and GDP tend to have lower child mortality and higher life expectancy
- Demographic Transition:** Lower fertility rates are strongly associated with higher life expectancy
- Development Clustering:** These correlations suggest distinct country groupings based on development levels

4. Outlier Detection Analysis

4.1 Univariate Outliers (IQR Method)

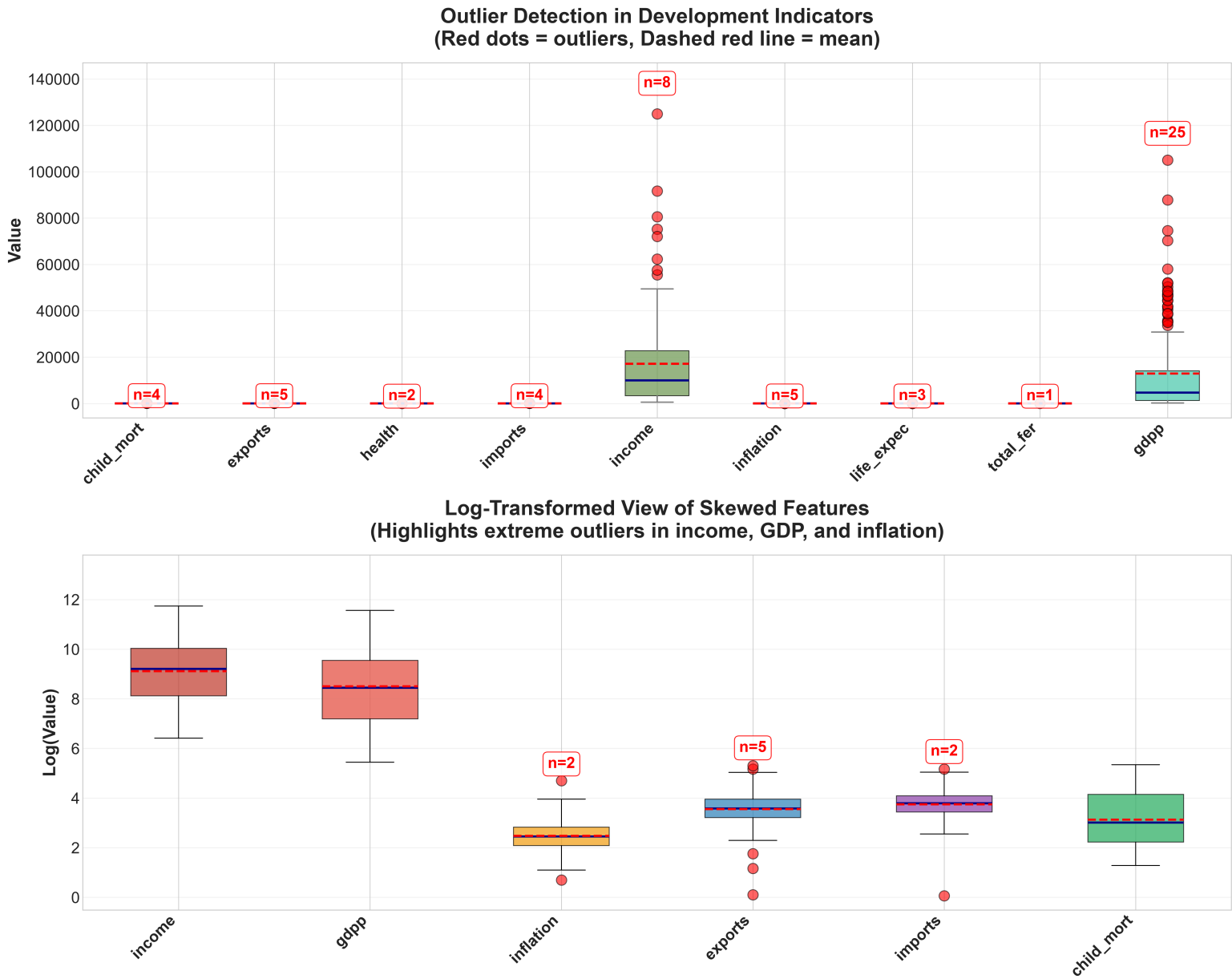


Figure 3: Outlier detection across all development indicators

Outlier Summary

Feature	Outlier Count	Outlier %	Extreme (z>3)	Severe (z>4)
gdpp	25	14.97%	4	2
income	8	4.79%	4	1
inflation	5	2.99%	2	1
exports	5	2.99%	3	3
child_mort	4	2.40%	2	1
imports	4	2.40%	3	2
life_expec	3	1.80%	1	1
health	2	1.20%	1	1
total_fer	1	0.60%	1	0
Total	57	3.79%	21	12

4.2 Multivariate Outliers (Isolation Forest)

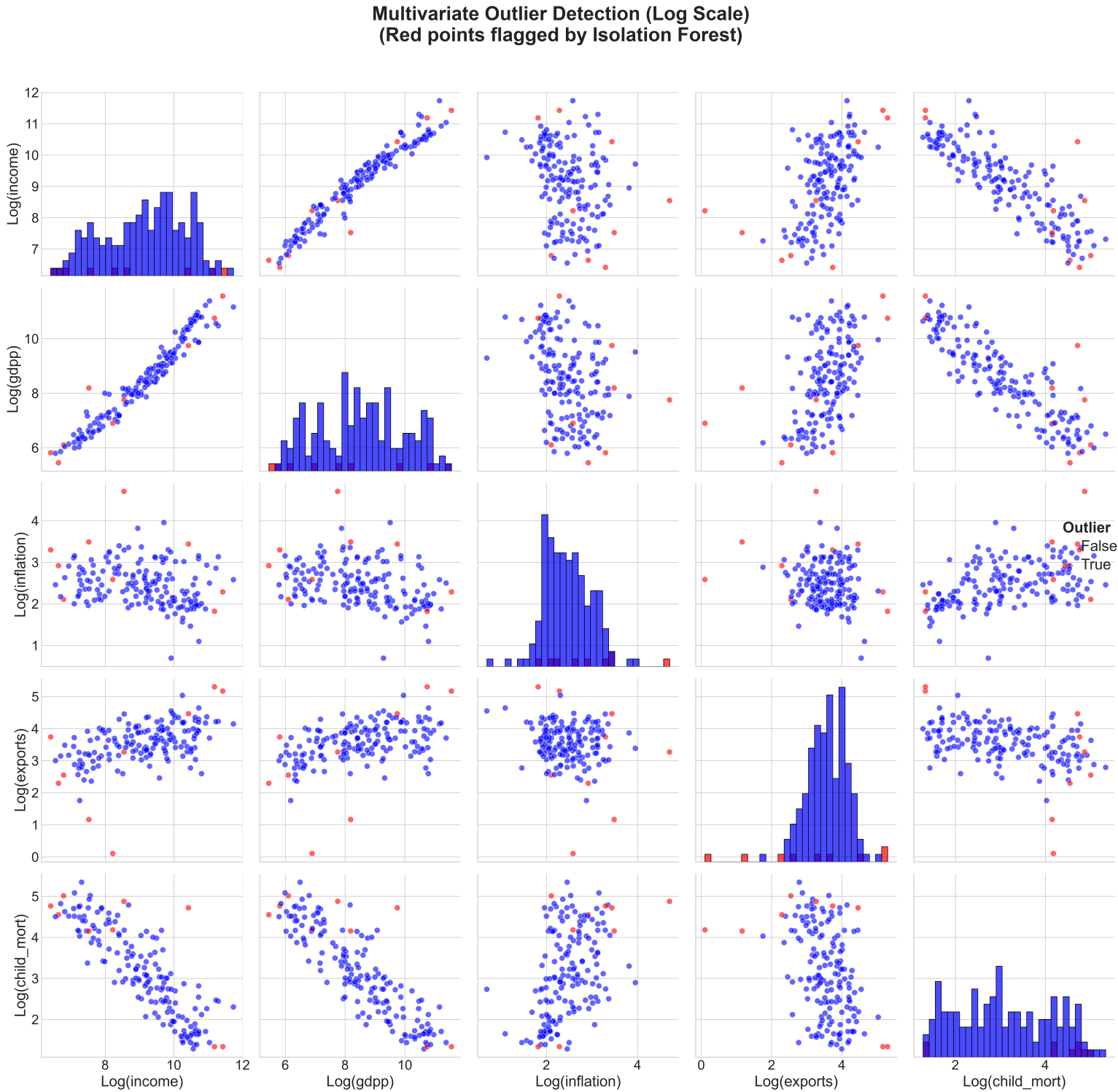


Figure 4: Multivariate outlier detection using Isolation Forest (log-transformed scale)

Multivariate Outliers Identified

#	Country	Key Indicators	Pattern
1	Burundi	Very low income (764), <i>lowGDP</i> (231)	Extreme poverty
2	Central African Republic	High child mortality (149), low income (\$888)	Extreme vulnerability
3	Congo, Dem. Rep.	Very low income (\$609), high inflation (20.8%)	Extreme poverty + inflation
4	Equatorial Guinea	High child mortality (111), high inflation (24.9%)	Resource-rich + health challenges
5	Luxembourg	Very high income (91, 700), <i>GDP</i> (105,000), exports (175%)	High-income outlier
6	Myanmar	Very low exports (0.11%), moderate child mortality (64.4)	Trade-isolated outlier
7	Nigeria	High inflation (104%), high child mortality (130)	Extreme inflation + health challenges
8	Singapore	Extremely high exports (200%), high income (\$72,100)	Trade-dependent outlier
9	Timor-Leste	High inflation (26.5%), moderate child mortality (62.6)	Inflation + development challenges

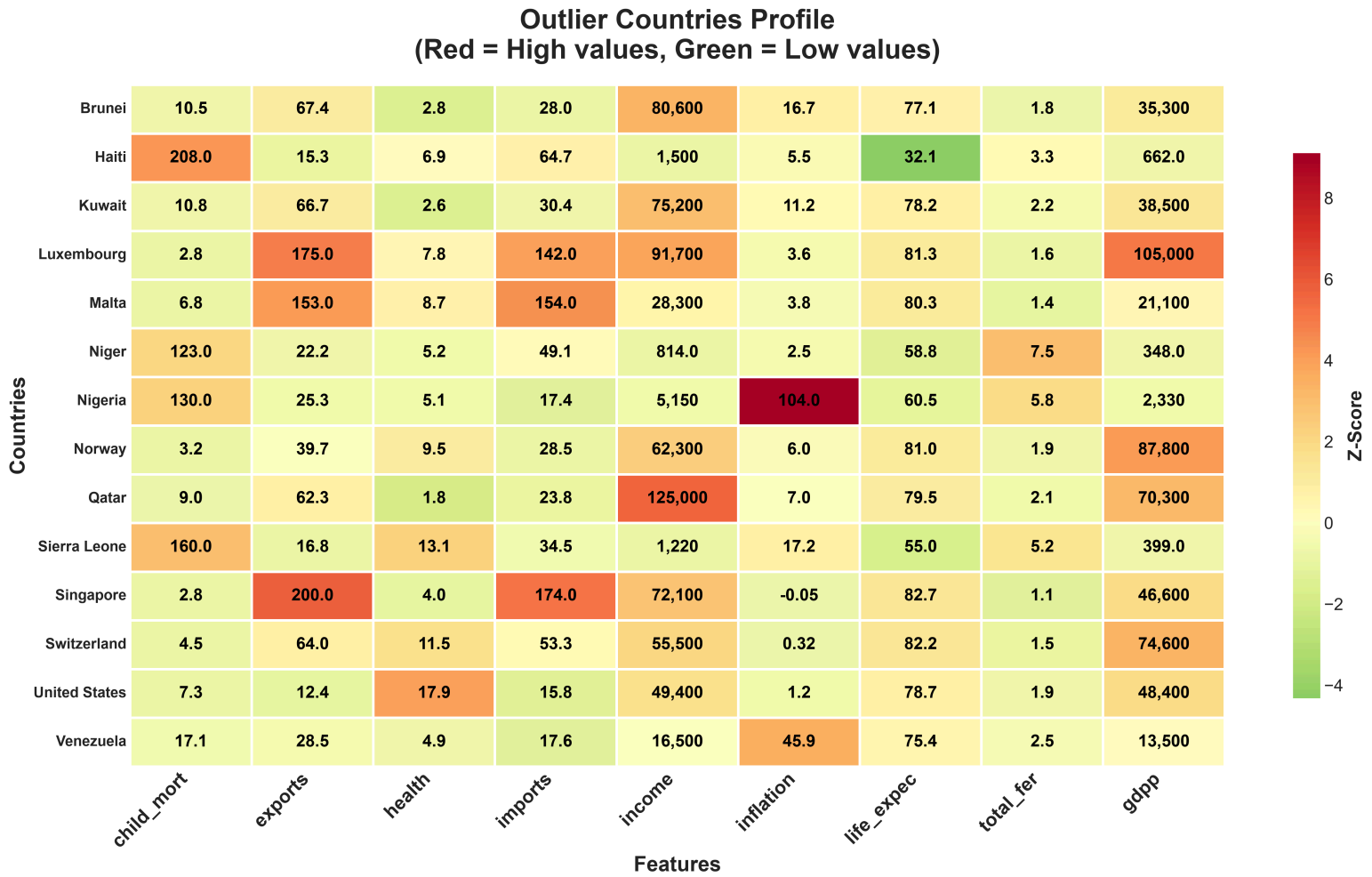


Figure 5: Outlier countries profile heatmap

Most Extreme Country: Nigeria

Feature	Nigeria Value	Global Mean	Difference
Inflation	104.00%	7.78%	+96.22% (z=4.76)
Child Mortality	130.00	38.27	+91.73
Total Fertility	5.84	2.95	+2.89
Income	\$5,150	\$17,144.69	-\$11,994.69
GDP per capita	\$2,330	\$12,964.16	-\$10,634.16

Note: Nigeria is the only country with a z-score > 4 for inflation (z=4.76), confirming it as the most extreme observation in the dataset.

Outlier Handling Decision

No observations were removed or capped because the extreme values represented plausible country-level socioeconomic conditions rather than confirmed data errors. Instead, the **Yeo-Johnson transformation** was used to reduce skewness and limit the disproportionate influence of highly skewed variables while preserving the observations.

5. Data Transformation Analysis

5.1 Feature Skewness Assessment

Initial skewness analysis identified significant skewness in multiple features, necessitating transformation.

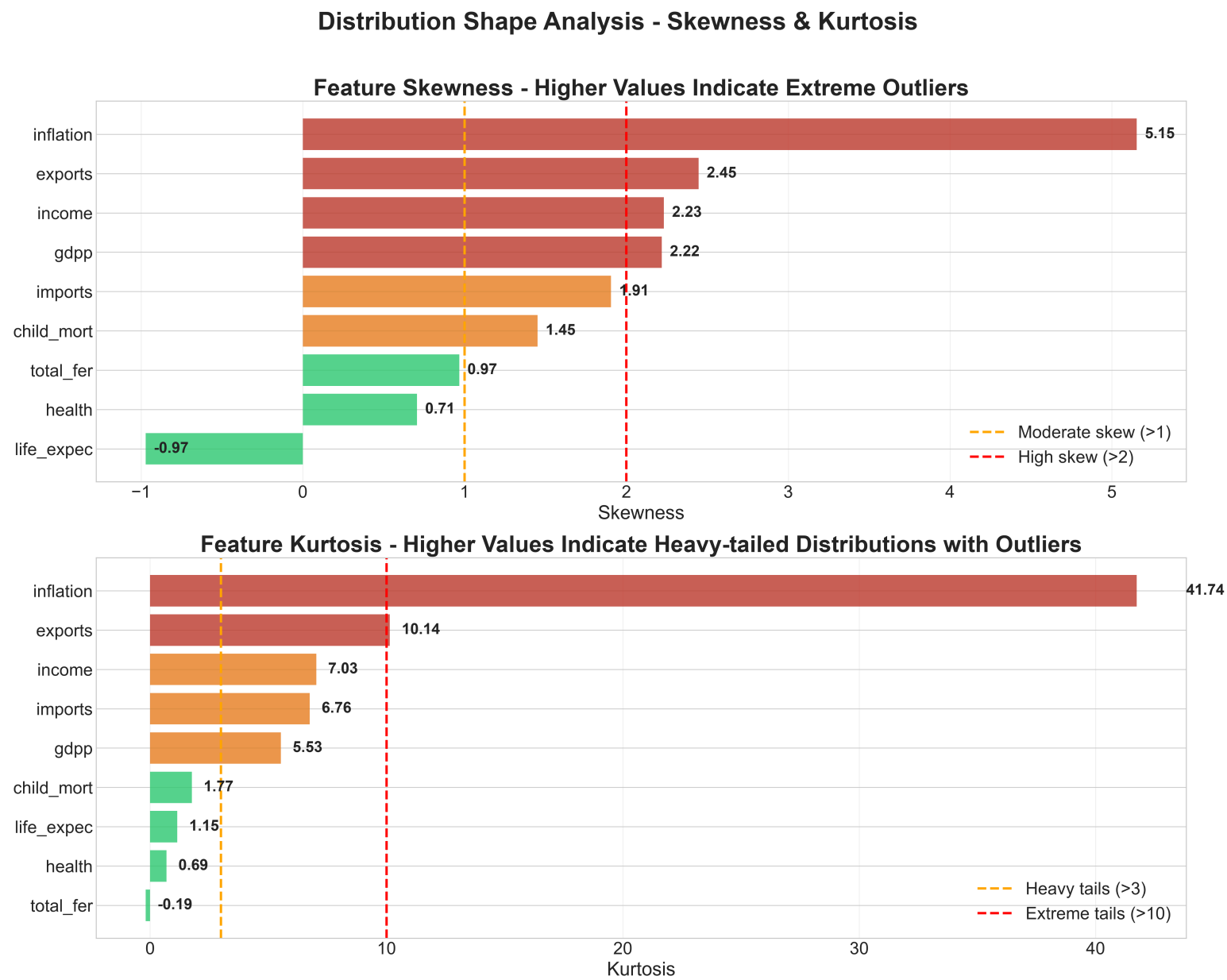


Figure 6: Skewness and kurtosis analysis of original features

Pre-Transformation Skewness

Feature	Skewness	Kurtosis	Interpretation
inflation	5.15	41.74	Extremely right-skewed
exports	2.45	10.14	Highly right-skewed
income	2.23	7.03	Highly right-skewed
gdpp	2.22	5.53	Highly right-skewed
imports	1.91	6.76	Moderately right-skewed
child_mort	1.45	1.77	Moderately right-skewed
total_fer	0.97	-0.19	Approximately symmetric
health	0.71	0.69	Approximately symmetric
life_expec	-0.97	1.15	Approximately symmetric

5.2 Transformation Selection

Yeo-Johnson transformation was selected over logarithmic transformation due to:

- 1. **Handling zero and negative values** (inflation has negative values)
- 2. **Superior performance** across all features
- 3. **Preservation of outlier information**

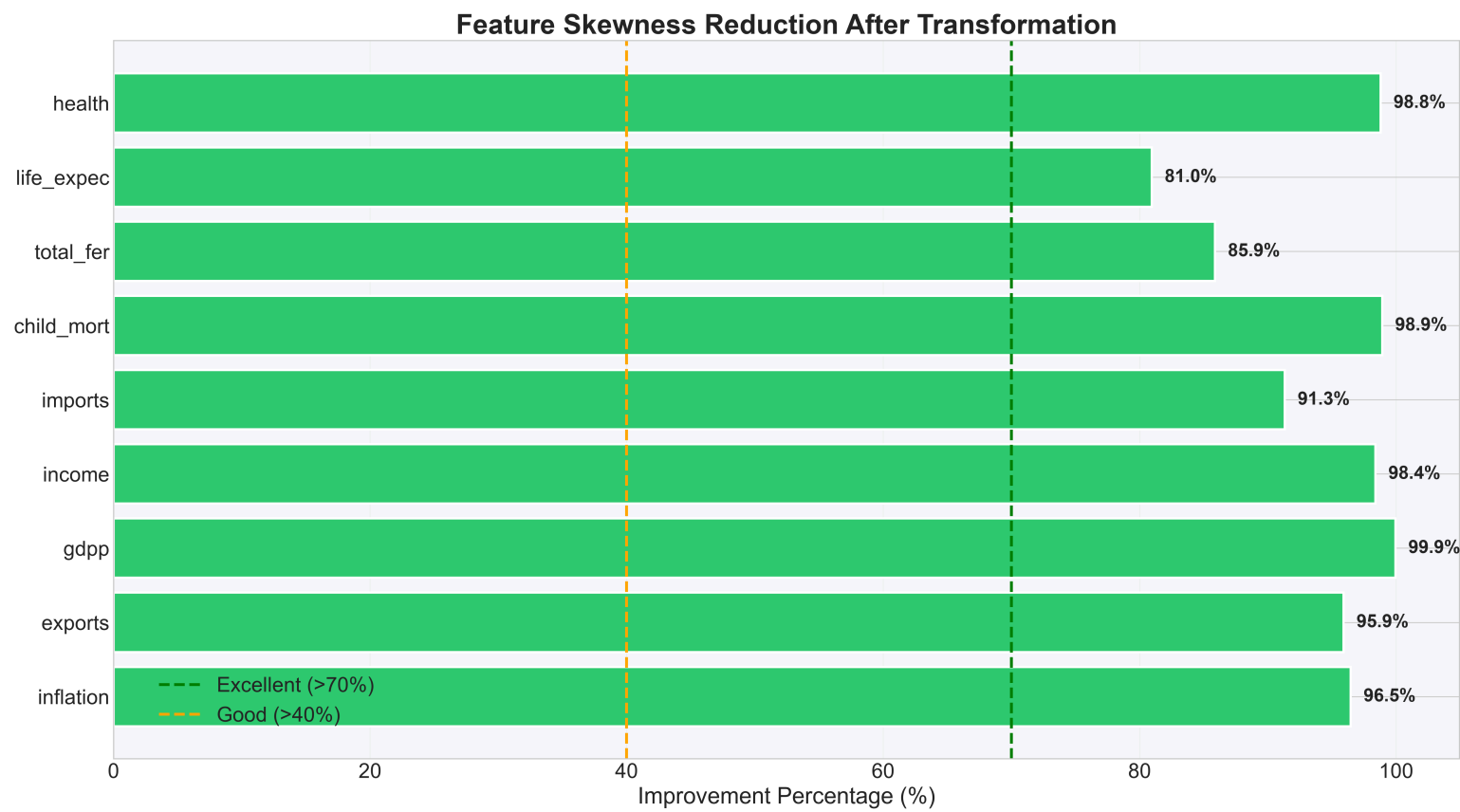


Figure 7: Transformation improvement by feature

Transformation Comparison Results

Feature	Original	Yeo-Johnson	Log	Improvement
child_mort	1.45	0.02	0.07	98.91%
exports	2.45	0.10	1.09	95.91%
health	0.71	-0.01	0.23	98.80%
imports	1.91	0.17	1.82	91.32%
income	2.23	-0.04	0.24	98.39%
inflation	5.15	0.18	0.45	96.46%
life_expec	-0.97	-0.18	1.63	80.95%
total_fer	0.97	0.14	0.54	85.88%
gdpp	2.22	0.00	0.01	99.95%

Total Improvement: Yeo-Johnson (94.1%) vs. Log (75.0%) average reduction

5.3 Before vs. After Comparison

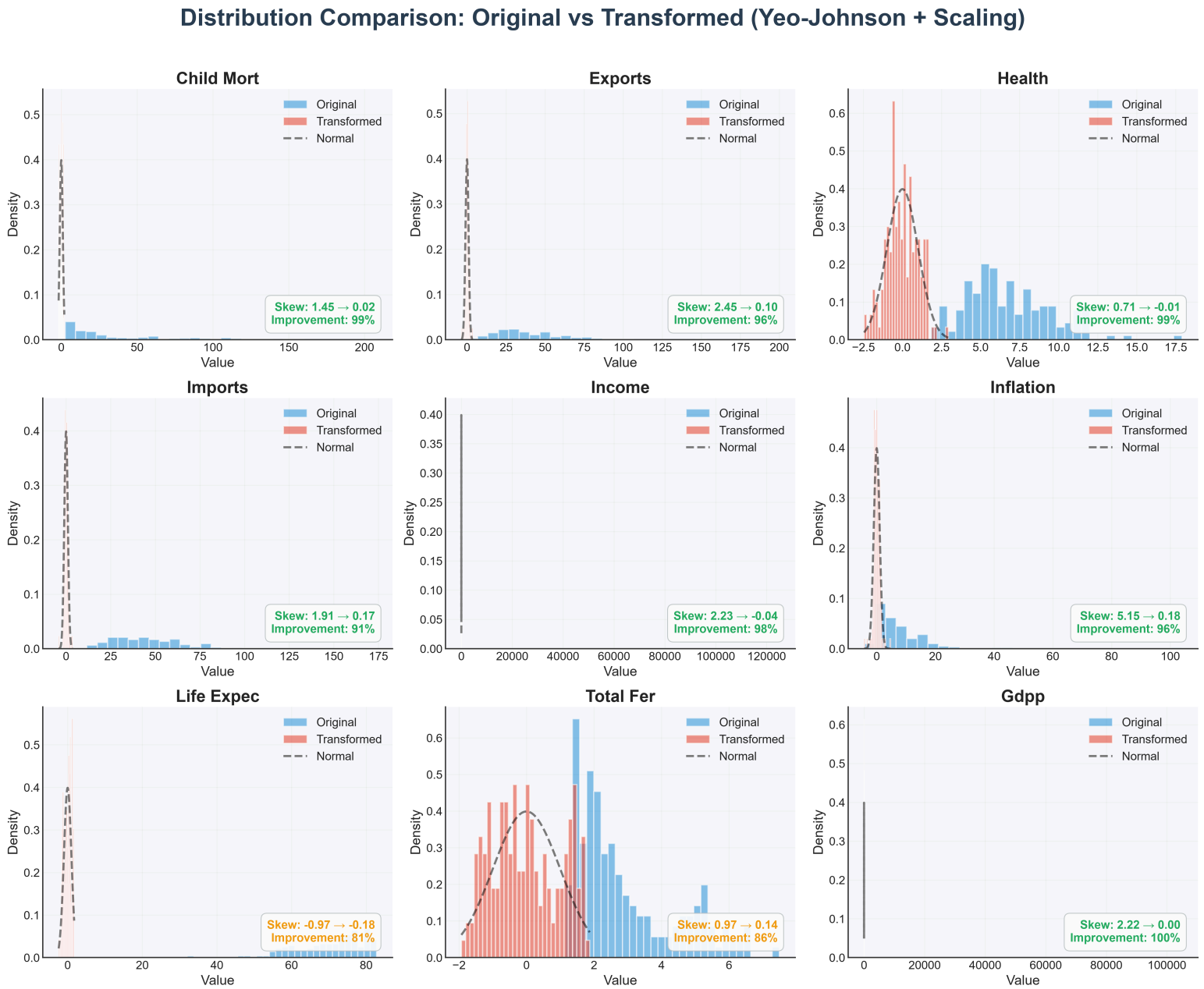


Figure 8: Comparison of original and transformed distributions

The transformation successfully:

- **Centered distributions** around the normal reference
- **Reduced skewness** from an average of 2.23 to 0.08
- **Preserved outliers** while reducing their leverage
- **Created approximately symmetric distributions** across all features

5.4 Normality Assessment

Feature	Skewness After Transform	Status
child_mort	0.016	✔ Excellent
exports	0.100	✔ Excellent
health	-0.009	✔ Excellent
imports	0.165	✔ Excellent
income	-0.036	✔ Excellent
inflation	0.183	✔ Excellent
life_expec	-0.185	✔ Excellent
total_fer	0.137	✔ Excellent
gdpp	0.001	✔ Excellent

6. Preprocessing Results

6.1 Final Preprocessed Data

The preprocessing pipeline successfully transformed the raw data into a clean, standardized dataset ready for clustering analysis.

Metric	Result
Original Data Shape	(167, 10)
Processed Data Shape	(167, 9)
Features Retained	9
Feature Names	child_mort, exports, health, imports, income, inflation, life_expec, total_fer, gdpp

6.2 Standardized Data Sample

Country	child_mort	exports	health	imports	income	inflation	life_expec	total_fer	gdpp
Afghanistan	1.27	-1.67	0.40	0.06	-1.37	0.46	-1.53	1.54	-1.46
Albania	-0.22	-0.38	0.03	0.23	0.01	-0.15	0.62	-0.99	-0.12
Algeria	0.22	0.11	-0.99	-0.61	0.23	1.06	0.65	0.31	-0.06
Angola	1.51	0.94	-1.73	-0.03	-0.41	1.51	-1.24	1.62	-0.22
Antigua and Barbuda	-0.64	0.39	-0.17	0.64	0.57	-0.72	0.70	-0.37	0.61

6.3 Verification Checks

The preprocessing pipeline was validated with the following checks:

Check	Expected	Actual	Status
Mean	≈ 0	0.0000	✔ Passed
Standard Deviation	≈ 1	1.0030	✔ Passed
Skewness	< 0.5	0.1850	✔ Passed

All verification checks passed successfully. The data is now properly standardized and ready for clustering analysis.

Summary Statistics After Transformation

Feature	Mean	Std	Skewness	Kurtosis
child_mort	0.00	1.003	0.016	-1.168
exports	0.00	1.003	0.100	1.684
health	0.00	1.003	-0.009	-0.258
imports	0.00	1.003	0.165	2.304
income	0.00	1.003	-0.036	-0.856
inflation	0.00	1.003	0.183	4.617
life_expec	0.00	1.003	-0.185	-0.981
total_fer	0.00	1.003	0.137	-1.139
gdpp	0.00	1.003	0.001	-0.956

7. Key Findings Summary

7.1 Data Quality

Metric	Result	Status
Total Observations	167 countries	✔
Total Features	9 development indicators	✔
Missing Values	0 (0%)	✔ Complete
Duplicate Rows	0	✔ Unique
Feature Coverage	100% complete	✔ Full

7.2 Distribution Characteristics

Characteristic	Details
Skewed Features	6 features with significant positive skewness (> 1.0)
Highest Skewness	Inflation (5.15), Exports (2.45), Income (2.23)
Strongest Correlations	child_mort ↔ life_expec (-0.887), income ↔ gdpp (+0.896)
Key Pattern	Strong relationships between health, demographic, and economic indicators

Features by Skewness Level

Skewness Level	Features
Extreme (>5.0)	Inflation (5.15)
High (2.0–3.0)	Exports (2.45), Income (2.23), GDP per capita (2.22)
Moderate (1.0–2.0)	Imports (1.91), Child mortality (1.45)
Symmetric (<1.0)	Total fertility (0.97), Health (0.71), Life expectancy (-0.97)

7.3 Outlier Insights

Metric	Result
Total Outliers Detected	57 across all features
Outlier Percentage	3.79% of total data
Most Outlier-Prone Feature	GDP per capita (15% outliers)
Most Extreme Observation	Nigeria (inflation: 4.76 standard deviations)
Multivariate Outliers	9 identified using Isolation Forest

Outlier Distribution by Feature

Feature	Outliers	Outlier %	Extreme (z>3)	Severe (z>4)
gdpp	25	14.97%	4	2
income	8	4.79%	4	1
inflation	5	2.99%	2	1
exports	5	2.99%	3	3
child_mort	4	2.40%	2	1
imports	4	2.40%	3	2
life_expec	3	1.80%	1	1
health	2	1.20%	1	1
total_fer	1	0.60%	1	0

7.4 Transformation Success

Metric	Before	After	Improvement
Average Skewness	2.23	0.08	96.4% ↓
Maximum Skewness	5.15	0.18	96.5% ↓
Features with Skewness < 0.2	0	9	100%

Feature-Level Transformation Results

Feature	Original Skewness	After Yeo-Johnson	Improvement
child_mort	1.45	0.02	98.9%
exports	2.45	0.10	95.9%
health	0.71	-0.01	98.8%
imports	1.91	0.17	91.3%
income	2.23	-0.04	98.4%
inflation	5.15	0.18	96.5%
life_expec	-0.97	-0.18	81.0%
total_fer	0.97	0.14	85.9%
gdpp	2.22	0.00	99.9%

Transformation Comparison:

Method	Average Reduction	Status
Yeo-Johnson	94.1%	✅ Superior
Logarithmic	75.0%	Acceptable
Yeo-Johnson Advantage	+19.1%	Better handling of zeros/negatives

✅ Outliers preserved while reducing their influence on the analysis

7.5 Preprocessing Results

Metric	Result	Status
Data Shape	167 × 9	✅
Mean	0.0000	✅ Centered
Standard Deviation	1.0030	✅ Scaled
Max Skewness	0.1850	✅ < 0.5 threshold
Features Retained	All 9 development indicators	✅
Readiness	Ready for clustering analysis	✅

7.6 Summary Dashboard

Category	Key Metric	Result
Data Quality	Completeness	100%
Distribution	Average Skewness	0.08 (96.4% improvement)
Outliers	Total Detected	57 (3.79%)
Transformation	Method	Yeo-Johnson (94.1% improvement)
Preprocessing	Standardization	mean=0, std≈1
Status	Clustering Readiness	✅ Ready

✅ All preprocessing steps completed successfully. The dataset is now ready for clustering analysis.

8. Clustering Analysis

8.1 Methodology

Building on the preprocessed data, **K-Means clustering** was applied to identify natural groupings of countries based on their development profiles. The number of clusters was evaluated using inertia, silhouette score, Calinski-Harabasz index, and Davies-Bouldin index.

8.2 Determining the Number of Clusters

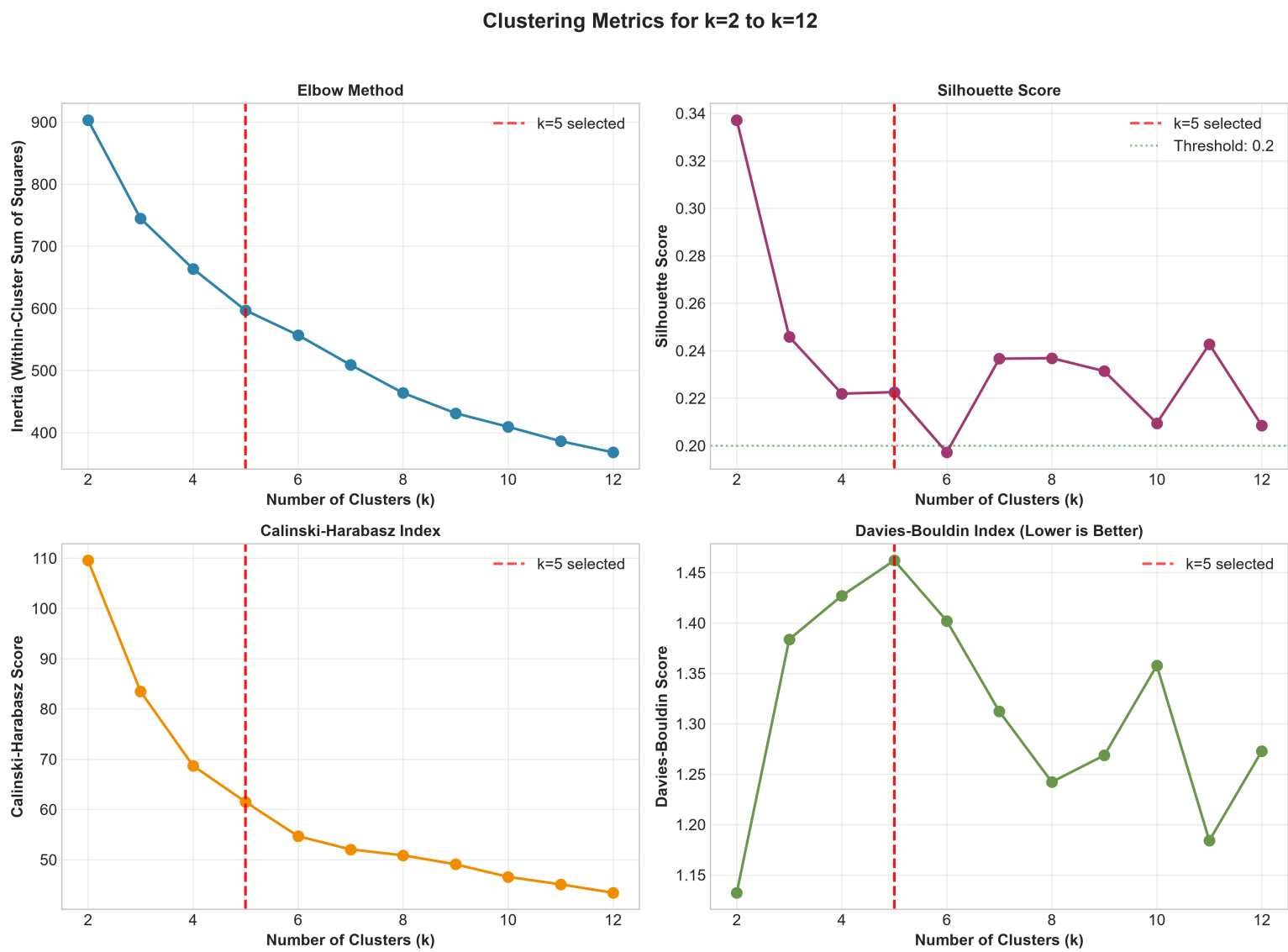


Figure 9: Clustering metrics evaluated for k = 2 to k = 12

Comprehensive Comparison (k = 3 to k = 7)

Metric	k=3	k=4	k=5	k=6	k=7
Silhouette Score	0.2458	0.2219	0.2226	0.1972	0.2366
Calinski-Harabasz	83.49	68.69	61.49	54.68	52.05
Davies-Bouldin	1.3837	1.4270	1.4620	1.4021	1.3123
Smallest Cluster	48	30	24	19	8
Size Ratio	1.33	1.90	1.83	2.00	4.75

8.3 Cluster Selection Rationale

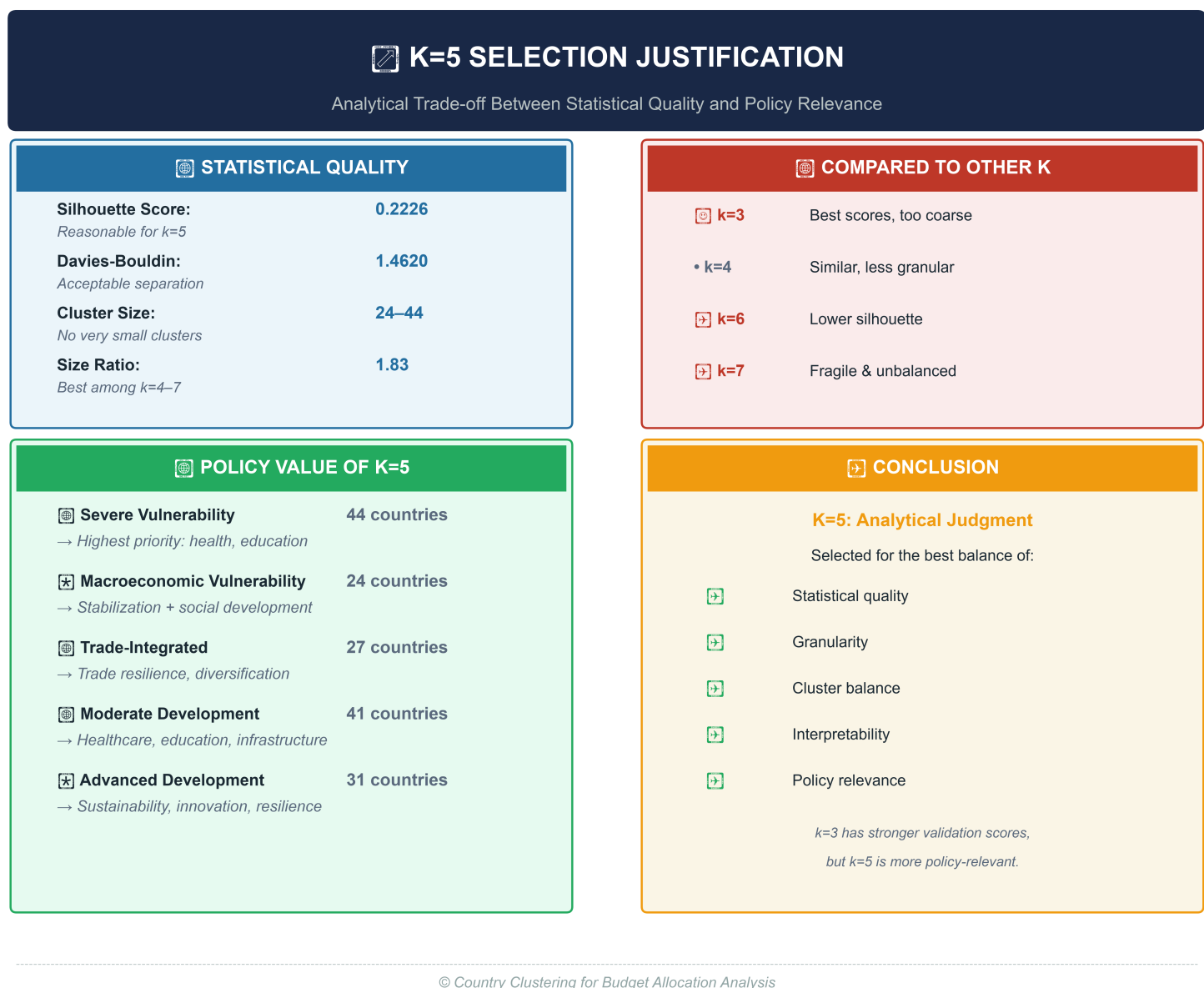


Figure 10: Rationale for selecting the five-cluster solution

The **k = 3 solution** provides the strongest overall internal clustering metrics. However, it produces a relatively broad segmentation and provides less differentiation among countries with distinct development characteristics.

The **k = 5 solution** provides greater analytical granularity while maintaining reasonable cluster sizes. It distinguishes several substantively different profiles, including high development vulnerability, trade-intensive development, intermediate development, higher-inflation development, and high development.

Therefore, $k = 5$ was selected as the final analytical solution based on a trade-off between statistical quality, cluster granularity, cluster size, interpretability, and policy relevance.

8.4 Final Cluster Distribution

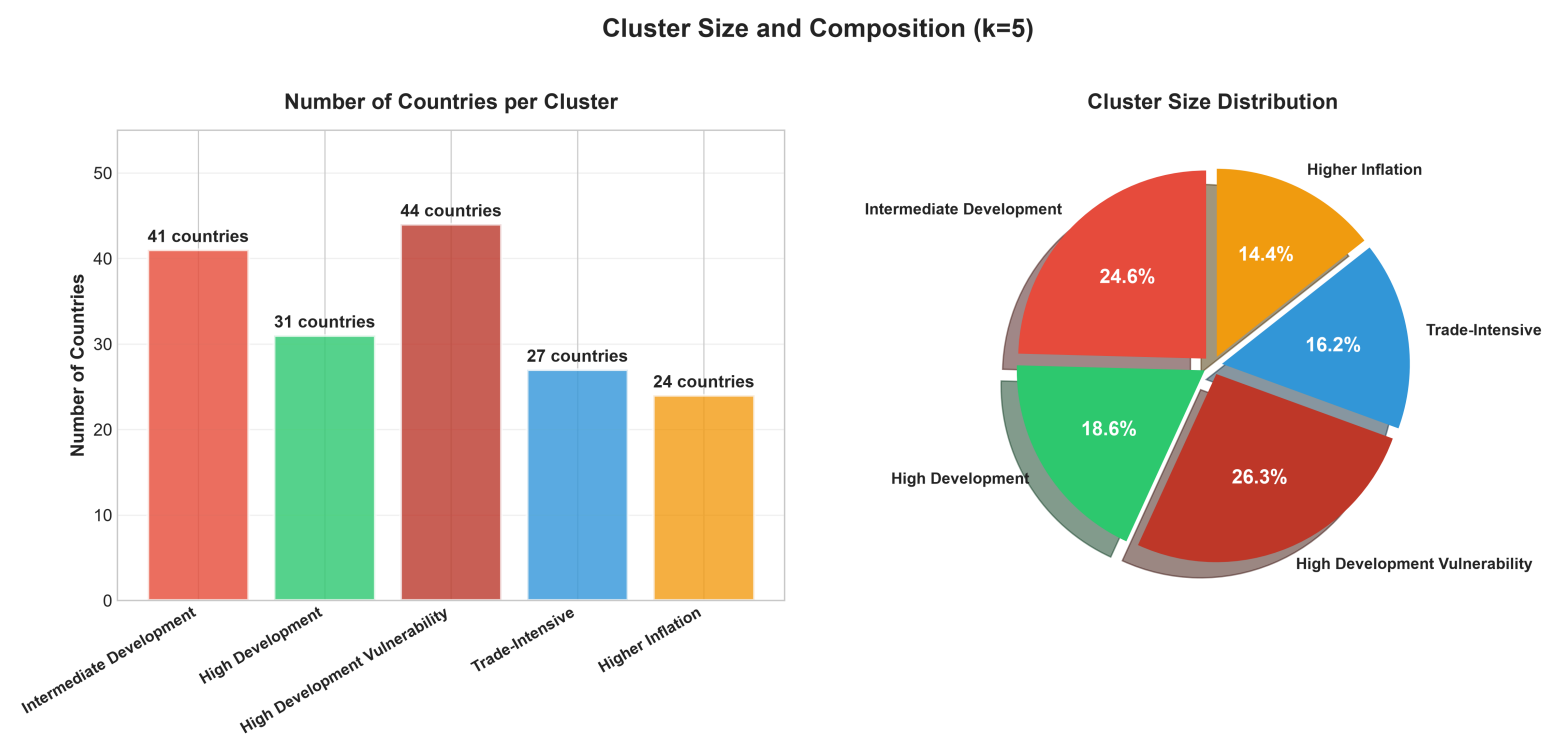


Figure 11: Distribution of countries across the five clusters

Cluster	Profile	Countries	Percentage
0	Intermediate Development	41	24.6%
1	High Development	31	18.6%
2	High Development Vulnerability	44	26.3%
3	Trade-Intensive Development	27	16.2%
4	Higher-Inflation Development	24	14.4%

8.5 PCA Visualization

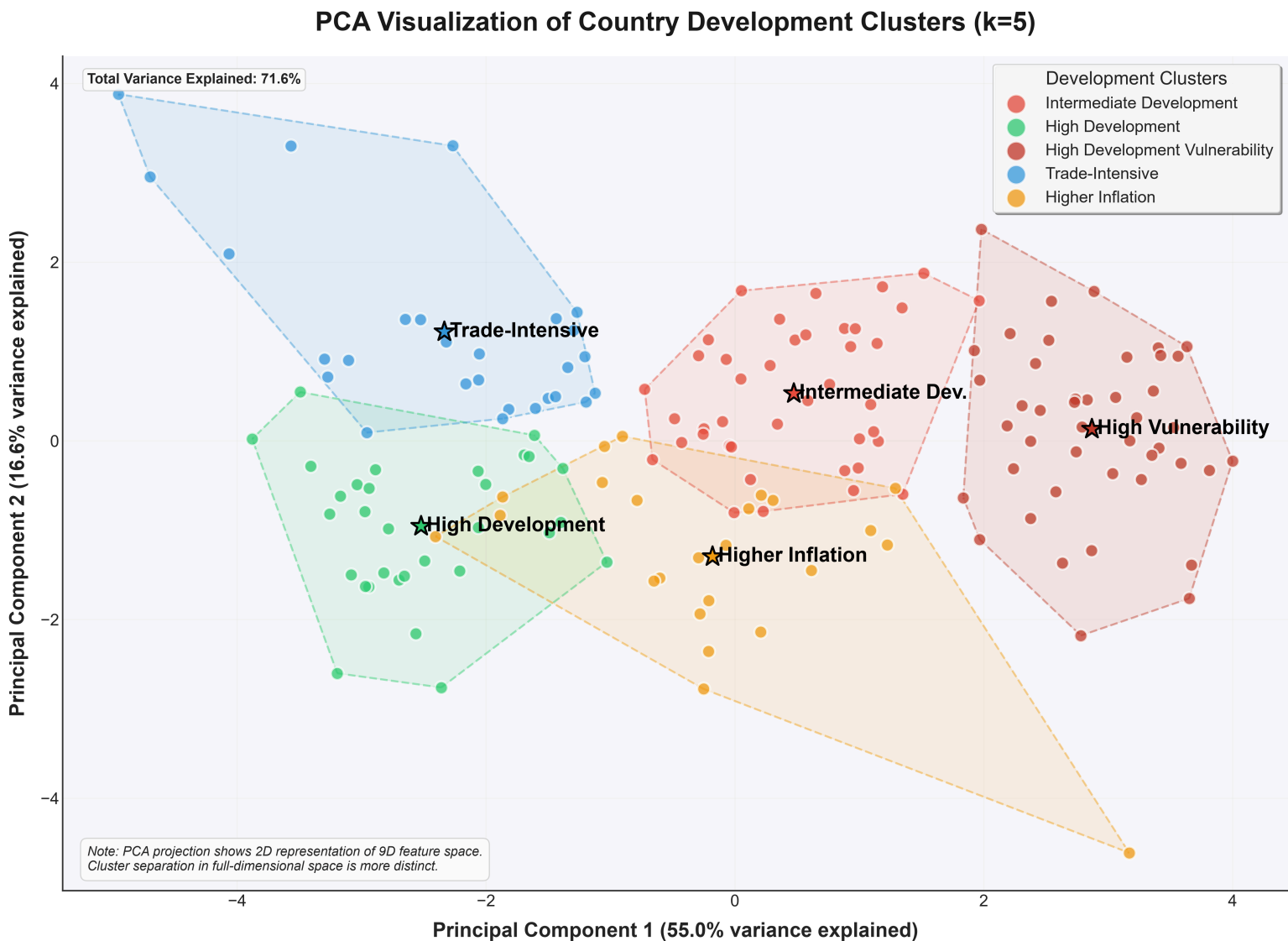


Figure 12: PCA visualization illustrating the structure and relative separation of the five clusters

Total variance explained: 71.6%

- PC1:** 55.0% - Development Gradient
- PC2:** 16.6% - Trade Orientation

PCA Interpretation

PC1 — Development Gradient: Higher PC1 scores are associated with higher child mortality and fertility and lower income, GDP per capita, and life expectancy, representing a gradient from stronger toward weaker socioeconomic development.

PC2 — Trade Orientation: Higher PC2 scores are associated with higher imports and exports, capturing the trade intensity dimension of development.

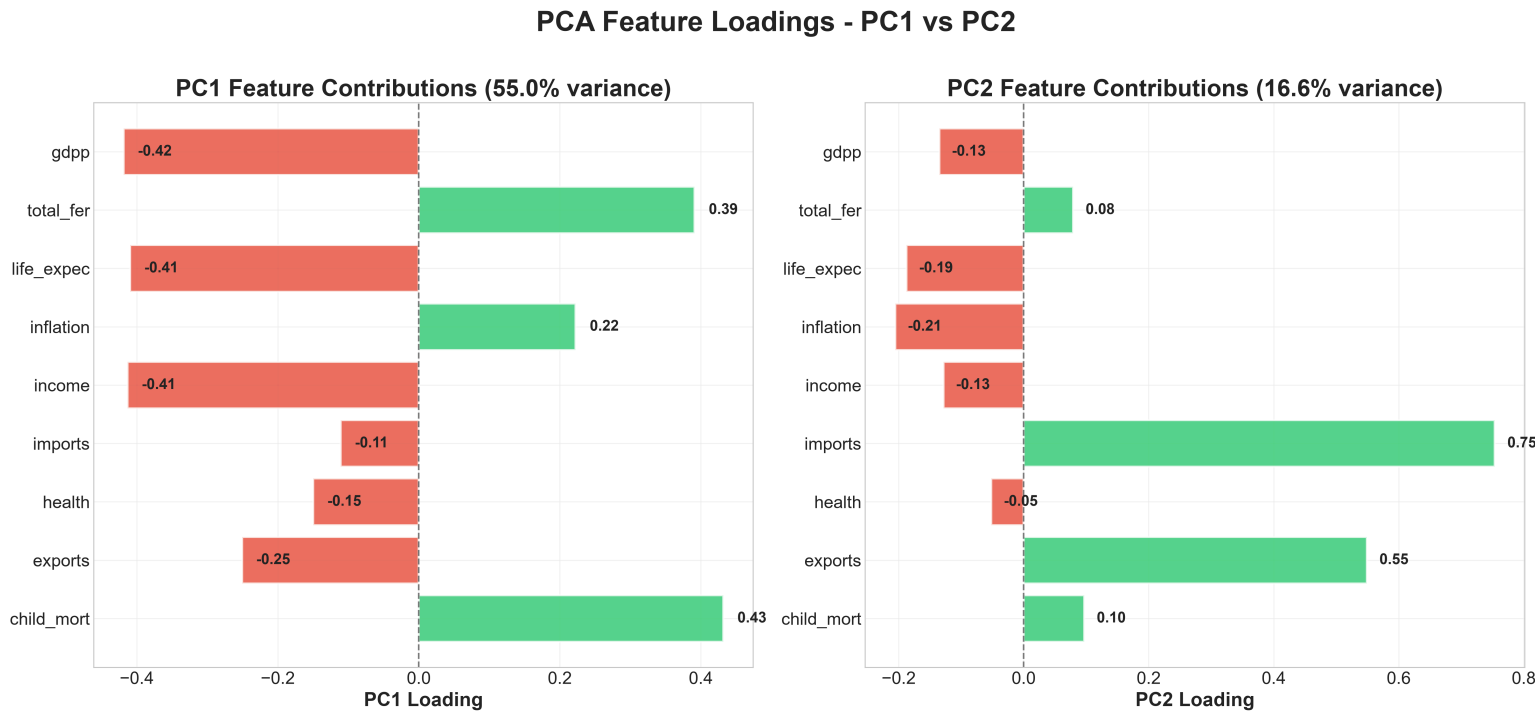


Figure 13: Feature loadings on the first two principal components

8.6 Cluster Profiles



Figure 14: Heatmap of standardized cluster means

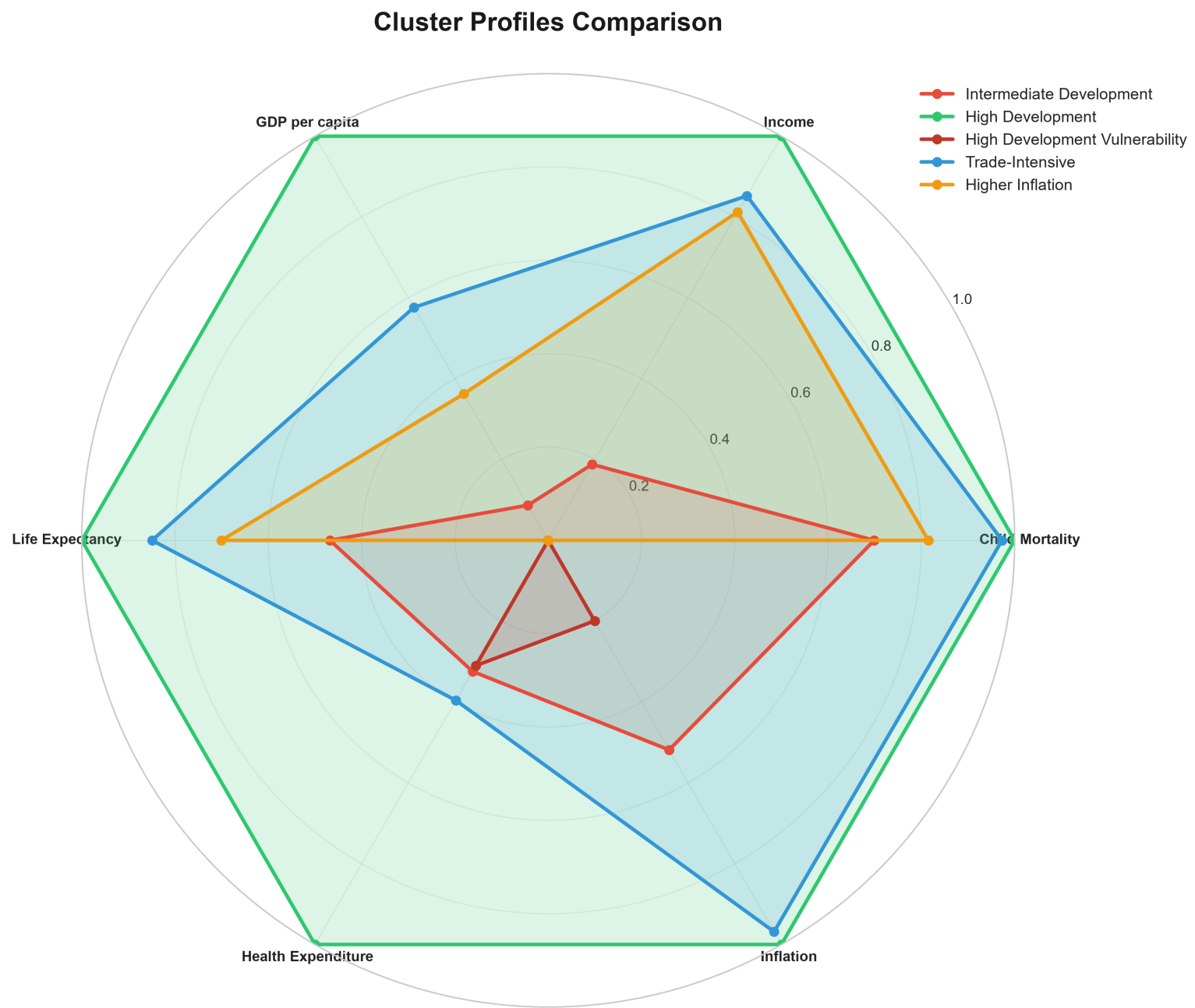


Figure 15: Comparison of the five cluster profiles across development indicators

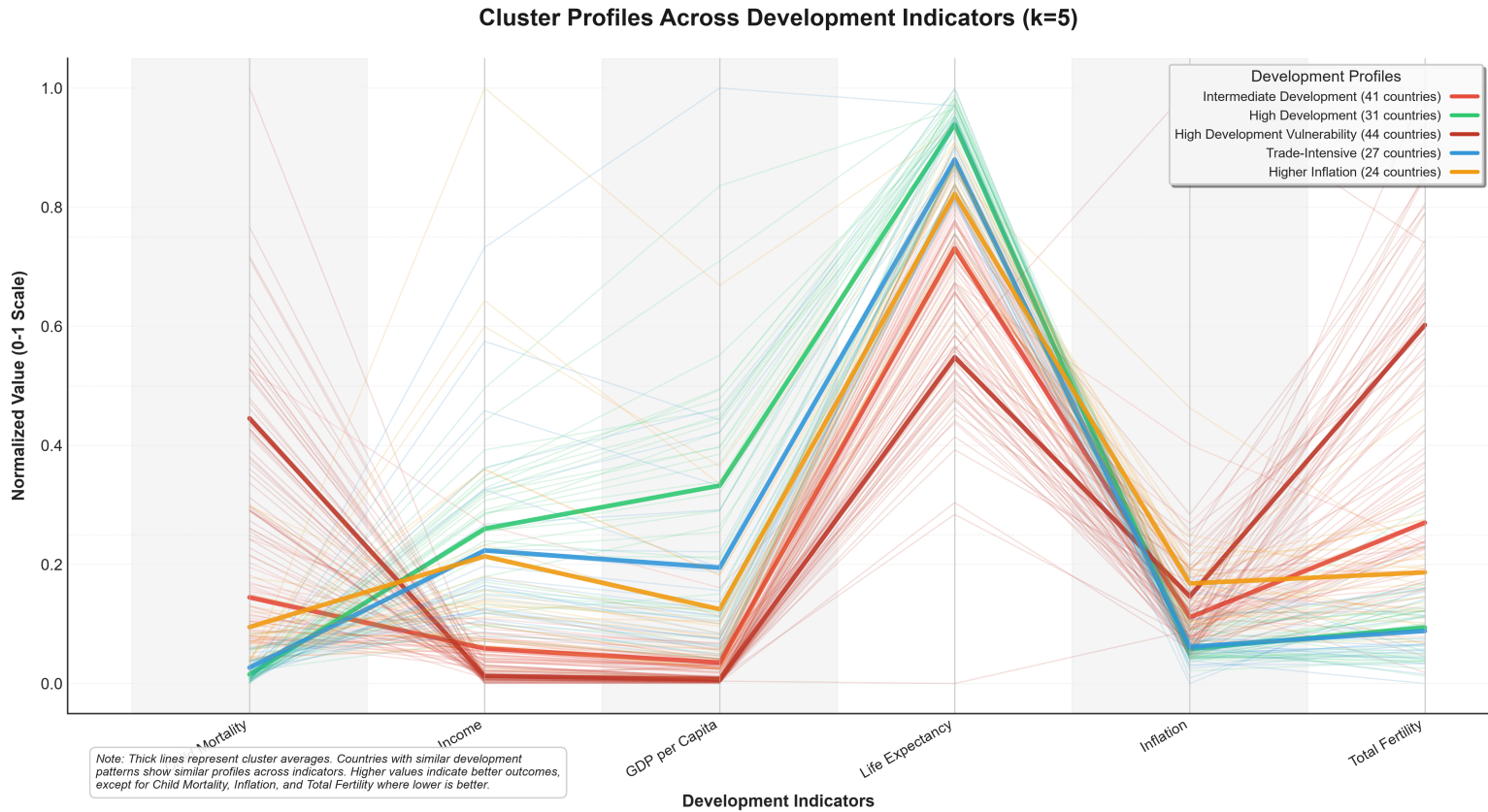


Figure 16: Parallel coordinates plot showing cluster profiles across all features

Cluster 0: Intermediate Development Profile — 41 countries

Indicator	Value	Interpretation
Child mortality	32.3	Moderate
Life expectancy	69.2 years	Moderate
Income (USD)	7,958	Moderate
GDP per capita (USD)	3,917	Moderate
Total fertility	2.86	Moderate

Characteristics: Countries with an intermediate development profile across the indicators included in the analysis. Compared with the high development vulnerability cluster, these countries exhibit substantially better health and economic outcomes, while remaining below the strongest-performing cluster on several development indicators. The profile suggests continued needs in healthcare, education, infrastructure, and broader economic development.

Policy Priority: High | Budget: 25% (\$250M)

Cluster 1: High Development Profile — 31 countries

Indicator	Value	Interpretation
Child mortality	5.7	Very Low
Life expectancy	79.7 years	High
Income (USD)	32,959	High
GDP per capita (USD)	35,055	High
Inflation	2.1%	Low

Characteristics: Countries with the strongest overall development profile among the five clusters. Within the scope of the nine indicators analyzed, these countries exhibit comparatively favorable health and economic outcomes. Policy emphasis may therefore shift toward sustainability, innovation, social protection, and long-term resilience rather than basic development needs.

Policy Priority: Very Low | Budget: 2% (\$20M)

Cluster 2: High Development Vulnerability — 44 countries

Indicator	Value	Interpretation
Child mortality	94.1	Extremely High
Life expectancy	59.9 years	Very Low
Income (USD)	2,170	Very Low
GDP per capita (USD)	952	Very Low
Total fertility	4.97	Very High

Characteristics: Countries with the highest level of development vulnerability in the dataset. These characteristics indicate substantial health, demographic, and economic development challenges and support the highest policy priority for basic healthcare, education, infrastructure, and poverty-reduction interventions.

Policy Priority: Very High | Budget: 45% (\$450M)

Cluster 3: Trade-Intensive Development Profile — 27 countries

Indicator	Value	Interpretation
Exports	77.5%	Very High
Imports	76.9%	Very High
Child mortality	8.1	Low
Income (USD)	28,415	High
Inflation	2.5%	Low

Characteristics: Countries characterized by exceptionally high trade intensity and relatively strong development outcomes. The defining feature of this cluster within the selected indicators is the combination of high trade intensity and relatively favorable development outcomes. Policy considerations may therefore include trade resilience, economic diversification, and sustainable participation in international markets.

Policy Priority: Medium | **Budget:** 8% (\$80M)

Cluster 4: Higher-Inflation Development Profile — 24 countries

Indicator	Value	Interpretation
Inflation	14.0%	High
Child mortality	22.1	Moderate
Life expectancy	73.8 years	Moderate
Income (USD)	27,186	High

Characteristics: Countries with relatively stronger income and human-development outcomes that are distinguished from the other clusters primarily by comparatively elevated inflation. The profile therefore suggests a combination of relatively stronger development outcomes with greater inflationary pressure. Policy considerations may include macroeconomic stabilization alongside continued investment in health and social development.

Policy Priority: High | **Budget:** 20% (\$200M)

8.7 Statistical Characterization

ANOVA and Kruskal-Wallis tests were conducted to characterize differences among the five resulting cluster profiles. **All nine indicators showed statistically significant differences across clusters (p < 0.001).**

Feature	ANOVA F	p-value	η²	Highest	Lowest
Child Mortality	225.35	<0.001	0.848	High Development Vulnerability	High Development
GDP per Capita	142.12	<0.001	0.778	High Development	High Development Vulnerability
Income	134.95	<0.001	0.769	High Development	High Development Vulnerability
Life Expectancy	116.53	<0.001	0.742	High Development	High Development Vulnerability
Total Fertility	100.84	<0.001	0.713	High Development Vulnerability	Trade-Intensive
Imports	38.02	<0.001	0.484	Trade-Intensive	Higher Inflation
Exports	27.59	<0.001	0.405	Trade-Intensive	High Development Vulnerability
Health Expenditure	21.89	<0.001	0.351	High Development	Higher Inflation
Inflation	20.42	<0.001	0.335	Higher Inflation	High Development

Child mortality, GDP per capita, income, life expectancy, and total fertility exhibited the **strongest between-cluster differences**, with η² values of 0.848, 0.778, 0.769, 0.742, and 0.713, respectively.

9. Geographic Distribution

9.1 Global Cluster Distribution

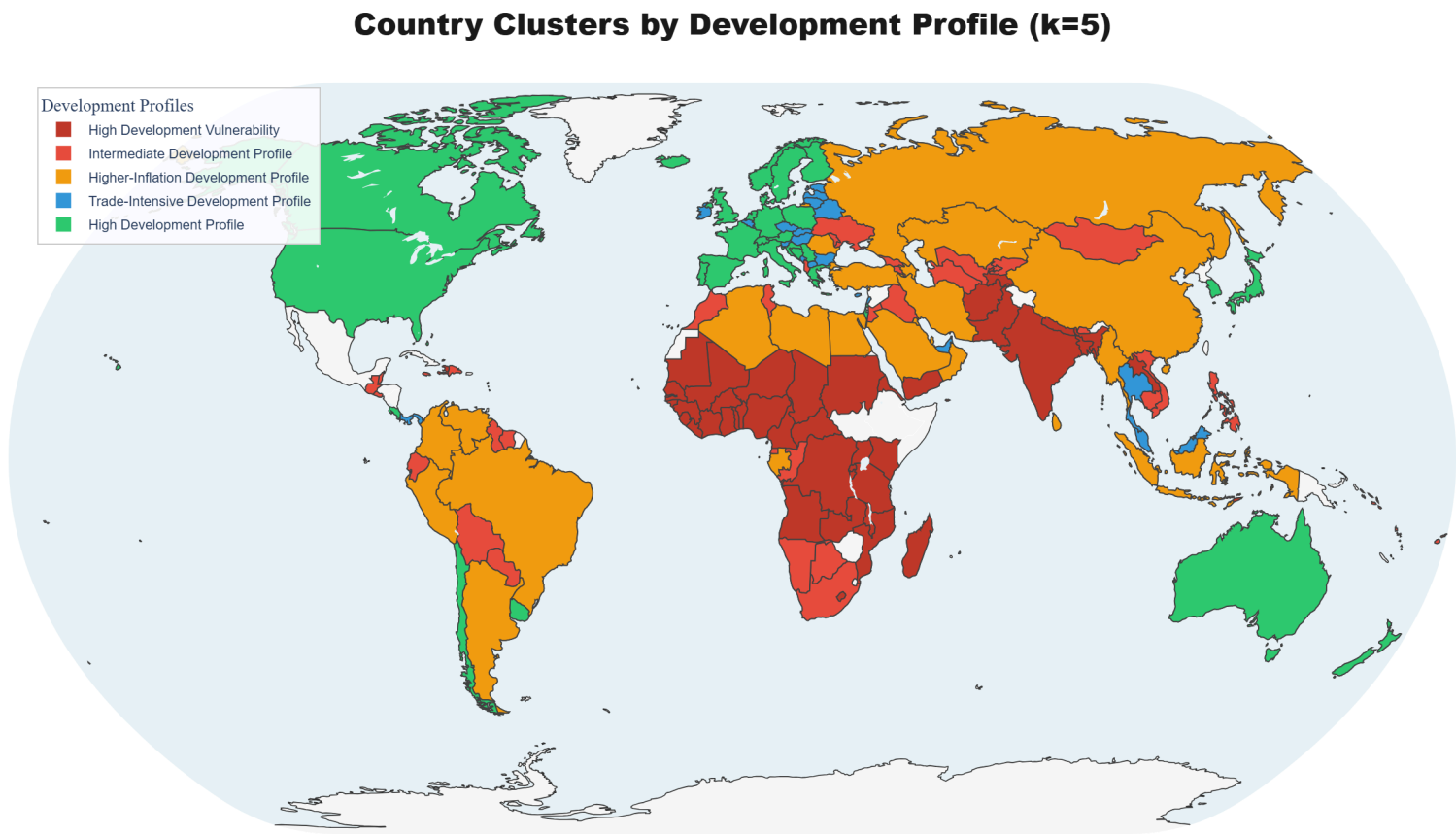


Figure 17: Geographic distribution of the five country clusters

10. Budget Allocation Strategy

10.1 Allocation Framework

The budget allocation represents an **illustrative need-based policy scenario** rather than a direct mathematical output of the clustering algorithm. The clustering model identifies development profiles and informs the relative allocation of the \$1 billion budget across clusters.

10.2 Allocation Overview

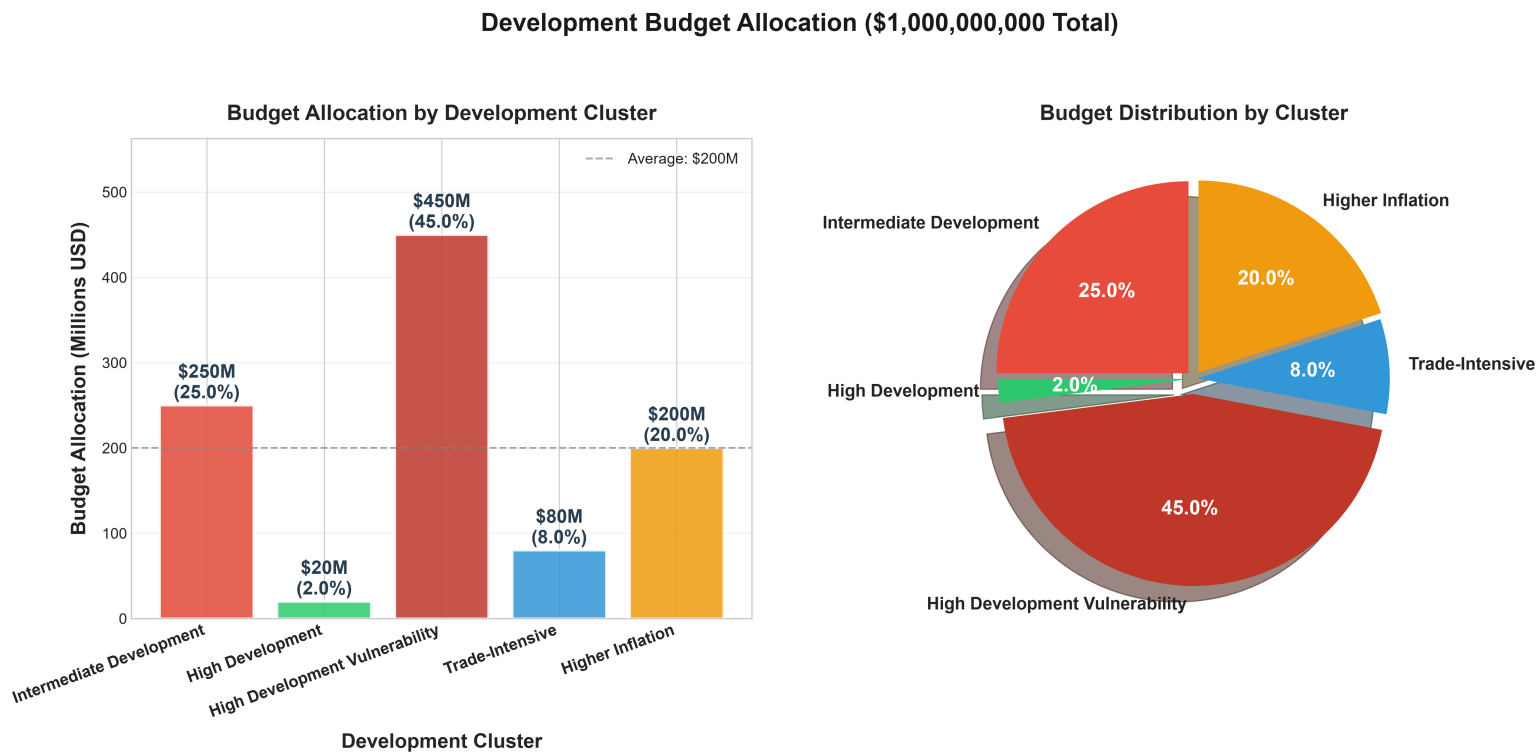


Figure 18: Illustrative budget allocation by cluster

10.3 Budget Allocation Summary

The following table summarizes the illustrative budget allocation across the five development clusters. The allocation follows a **needs-based approach**, prioritizing clusters with the most severe development challenges.

Cluster	Priority	Countries	Budget %	Amount (M)	Per Country
High Development Vulnerability	Very High	44	45%	\$450.0	\$10,227,273
Intermediate Development	High	41	25%	\$250.0	\$6,097,561
Higher-Inflation Development	High	24	20%	\$200.0	\$8,333,333
Trade-Intensive Development	Medium	27	8%	\$80.0	\$2,962,963
High Development	Very Low	31	2%	\$20.0	\$645,161
Total		167	100%	\$1,000.0	

Key Observations

Observation	Detail
Largest Allocation	High Development Vulnerability receives 45% (\$450M)
Smallest Allocation	High Development receives 2% (\$20M)
Highest Per Country	High Development Vulnerability: \$10.2M per country
Lowest Per Country	High Development: \$0.6M per country
Budget Ratio (Highest:Lowest)	22.5:1 (reflects needs-based prioritization)

Allocation Rationale by Cluster

Cluster	Budget Share	Rationale
High Development Vulnerability	45%	Most severe health, demographic, and economic challenges; highest priority for basic development and poverty-reduction interventions
Intermediate Development	25%	Substantial development gaps remain in health, income, and socioeconomic conditions
Higher-Inflation Development	20%	Relatively strong development indicators but substantial macroeconomic vulnerability, particularly high inflation and lower health expenditure
Trade-Intensive Development	8%	Relatively strong development outcomes with very high trade integration; emphasis on resilience, diversification, and sustainable integration
High Development	2%	Strongest overall human-development and socioeconomic profile; lowest development-assistance priority

10.4 Budget Flow Sankey Diagram

Budget Allocation Flow: \$1 Billion from Budget to Development Priorities

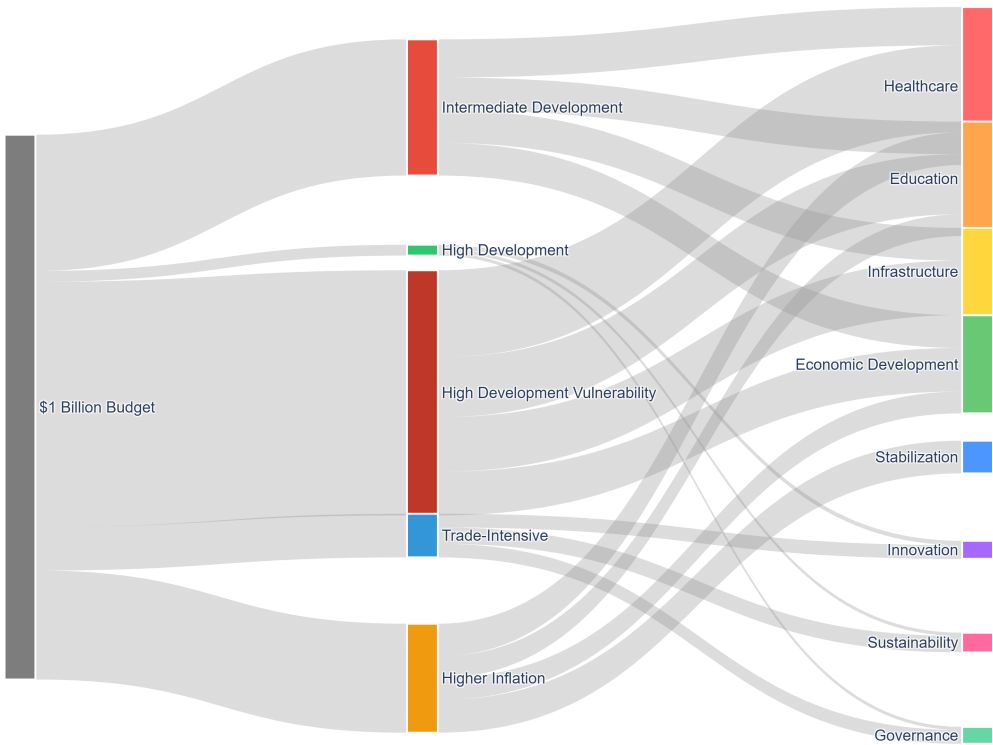


Figure 19: Illustrative budget flow from country clusters to policy areas

The Sankey diagram above visualizes the complete budget flow from the \$1 billion total budget through the five development clusters and into eight policy areas. This flow illustrates how cluster-level allocations translate into sectoral investments.

Flow Interpretation:

- **Left to Right:** Budget flows from the total budget → clusters → policy areas
- **Line Thickness:** Represents the relative size of each allocation
- **Color Coding:** Matches the cluster color scheme used throughout this report

10.5 Policy Area Breakdown

The cluster budgets are further allocated across eight policy areas based on the specific needs of each development profile. Healthcare and education receive the largest shares, reflecting their importance across all clusters.

Policy Area	Amount (M)	Share	Priority Clusters
Healthcare	\$290.0M	29.0%	All clusters (highest in vulnerability)
Education	\$210.0M	21.0%	Vulnerability + Intermediate
Economic Development	\$180.0M	18.0%	Vulnerability + Intermediate
Infrastructure	\$160.0M	16.0%	Vulnerability + Intermediate
Stabilization	\$60.0M	6.0%	Higher-Inflation
Sustainability	\$36.0M	3.6%	High + Trade-Intensive
Innovation	\$33.0M	3.3%	High + Trade-Intensive
Governance	\$31.0M	3.1%	High + Trade-Intensive
Total	\$1,000.0M	100%	

Policy Area Allocation by Cluster

Cluster	Healthcare	Education	Infrastructure	Economic Dev.	Stabilization	Innovation	Sustainability	Governance
High Development Vulnerability	\$160M	\$110M	\$100M	\$80M	—	—	—	—
Intermediate Development	\$70M	\$60M	\$60M	\$60M	—	—	—	—
Higher-Inflation Development	\$60M	\$40M	—	\$40M	\$60M	—	—	—
Trade-Intensive Development	—	—	—	—	—	\$25M	\$30M	\$25M
High Development	—	—	—	—	—	\$8M	\$6M	\$6M

Key Policy Insights

Insight	Detail
Healthcare Priority	Largest single policy area (29%) - critical for vulnerability clusters
Education Investment	Second largest (21%) - essential for long-term development
Stabilization Focus	Targeted exclusively at Higher-Inflation cluster (6%)
Sustainability & Innovation	Focused on High and Trade-Intensive clusters (7%)
Needs-Based Distribution	Vulnerability clusters receive basic services; advanced clusters receive forward-looking investments

10.6 Budget Allocation Summary Dashboard

Metric	Value
Total Budget	\$1,000,000,000
Clusters Funded	5
Countries Supported	167
Policy Areas	8
Largest Recipient	High Development Vulnerability (45%)
Smallest Recipient	High Development (2%)
Highest Per Country	High Development Vulnerability (\$10.2M)
Lowest Per Country	High Development (\$0.6M)

10.7 Budget Allocation Caveats

Important: This budget allocation represents an **illustrative need-based policy scenario** rather than an empirically optimized funding formula. The clustering algorithm identifies country groupings; the budget percentages represent policy-planning assumptions informed by those groupings. The allocation should therefore be interpreted as a **framework for discussion and refinement** rather than a prescriptive allocation. Stakeholders are encouraged to adjust percentages based on additional policy considerations, country-specific contexts, and available resources.

11. Policy Recommendations

11.1 Cluster-Specific Recommendations

Cluster	Priority	Recommended Interventions
High Development Vulnerability	Very High	Emergency healthcare, child survival programs, poverty reduction, basic education, infrastructure development, economic stabilization
Intermediate Development	High	Healthcare system strengthening, education investment, infrastructure development, economic growth promotion, social protection
Higher-Inflation Development	High	Macroeconomic stabilization, monetary policy reform, healthcare investment, social protection, economic diversification
Trade-Intensive Development	Medium	Trade resilience, economic diversification, sustainable integration, innovation support, competitiveness enhancement
High Development	Very Low	Sustainable development, innovation, social protection, long-term resilience, knowledge economy development

11.2 Cross-Cutting Recommendations

- Healthcare:** Prioritize child survival programs in vulnerability clusters, universal health coverage in intermediate clusters, and health system innovation in advanced clusters
- Education:** Focus on basic education and literacy in vulnerable clusters, secondary education in intermediate clusters, and higher education and skills development in advanced clusters
- Economic Development:** Emphasis on poverty reduction and job creation in vulnerable clusters, industrialization in intermediate clusters, and innovation and competitiveness in advanced clusters
- Infrastructure:** Basic infrastructure (water, sanitation, energy) in vulnerable clusters, digital infrastructure in intermediate clusters, and sustainable infrastructure in advanced clusters

12. Limitations and Caveats

12.1 Methodological Caveats

- Clustering interpretation:** The five clusters represent data-driven profiles rather than objectively defined development categories. Cluster labels are interpretive and should not be interpreted as official country classifications.
- Internal validation:** The k = 5 solution does not maximize every internal clustering metric. The k = 3 solution produces stronger statistical scores on several metrics. Therefore, k = 5 was selected through a broader analytical judgment incorporating granularity, cluster size, interpretability, and policy relevance.
- Budget allocation:** The budget allocation is an **illustrative policy scenario**, not an empirically optimized allocation formula. The clustering algorithm identifies country groupings; the budget percentages represent policy-planning assumptions informed by those groupings.
- Statistical characterization:** ANOVA and Kruskal-Wallis tests demonstrate significant differences among the resulting clusters but should not be interpreted as independent evidence that the clustering solution is objectively optimal because the same indicators were used to construct the clusters.
- Cross-sectional analysis:** The analysis is based on country-level observations and does not capture changes in development indicators over time.

12.2 Future Work

- Temporal analysis:** Examine whether countries transition between development profiles over time
- External validation:** Compare the resulting profiles with established international development classifications
- Policy impact assessment:** Evaluate the effectiveness of needs-based allocation strategies
- Alternative clustering methods:** Compare the K-Means solution with additional clustering algorithms
- Sensitivity analysis:** Examine how different preprocessing and feature-selection choices affect cluster membership

13. Key Findings & Insights

13.1 Major Findings

- Five distinct socioeconomic profiles** were identified across 167 countries
- Child mortality, total fertility, life expectancy, GDP per capita, and income** are the strongest discriminating characteristics between the resulting clusters
- 44 countries (26.3%)** belong to the High Development Vulnerability profile and exhibit the most adverse combination of health, demographic, and economic indicators
- 31 countries (18.6%)** belong to the High Development Profile and exhibit the strongest overall socioeconomic outcomes in the dataset
- The illustrative budget allocation follows a **needs-based approach**, assigning the largest share to the High Development Vulnerability profile and smaller shares to stronger overall profiles
- Trade intensity and inflation** distinguish additional socioeconomic profiles beyond a simple development ranking

7. **All nine features** show statistically significant differences across the resulting clusters (p < 0.001)

13.2 Final Results at a Glance

Metric	Result
Countries analyzed	167
Development indicators	9
Selected clusters	5
Algorithm	K-Means
Silhouette score	0.2226
PCA variance explained	71.6%
Strongest ANOVA effect	Child mortality ($\eta^2 = 0.848$)
Illustrative budget	\$1B
Most vulnerable countries	44 (26.3%)
Advanced countries	31 (18.6%)

14. Conclusion

This comprehensive analysis has identified **five distinct country development profiles** from 167 countries using 9 development indicators. The analysis demonstrates that countries can be meaningfully grouped based on their multidimensional socioeconomic characteristics, with health, demographic, and economic indicators providing the strongest differentiation.

The selected k = 5 solution represents a **trade-off between statistical clustering quality and practical analytical usefulness** rather than a universally optimal number of clusters. The k = 3 solution provides stronger internal validation scores but offers less policy-relevant granularity.

The **illustrative \$1 billion budget allocation** demonstrates how clustering insights can inform resource allocation, prioritizing the most development-vulnerable profiles while allocating smaller shares to countries with stronger overall outcomes.

This analysis provides a foundation for policy discussion, resource allocation planning, and stakeholder engagement. The results should be interpreted as data-driven profiles that can inform decision-making rather than as definitive country classifications.

15. Appendix

15.1 Cluster Color Legend

Cluster	Name	Color	Hex	Count
0	Intermediate Develop...	■ Red	#E74C3C	41
1	High Development	■ Green	#2ECC71	31
2	High Development Vul...	■ Dark Red	#C0392B	44
3	Trade-Intensive	■ Blue	#3498DB	27
4	Higher Inflation	■ Orange	#F39C12	24

Figure 20: Cluster color legend used throughout the visualizations

15.2 Transformation Comparison Details

Feature	Original	Yeo-Johnson	Log	Yeo Better?
child_mort	1.45	0.02	0.07	✔ YES
exports	2.45	0.10	1.09	✔ YES
health	0.71	-0.01	0.23	✔ YES
imports	1.91	0.17	1.82	✔ YES
income	2.23	-0.04	0.24	✔ YES
inflation	5.15	0.18	0.45	✔ YES
life_expec	-0.97	-0.18	1.63	✔ YES
total_fer	0.97	0.14	0.54	✔ YES
gdpp	2.22	0.00	0.01	✔ YES

Total Improvement: Yeo-Johnson: 17.22, Log: 11.98 **Yeo-Johnson is 43.7% better!**

15.3 Complete ANOVA Results

Feature	ANOVA F	p-value	η^2	Effect Size
Child Mortality	225.35	<0.001	0.848	Very Large
GDP per Capita	142.12	<0.001	0.778	Very Large
Income	134.95	<0.001	0.769	Very Large
Life Expectancy	116.53	<0.001	0.742	Very Large
Total Fertility	100.84	<0.001	0.713	Very Large
Imports	38.02	<0.001	0.484	Large
Exports	27.59	<0.001	0.405	Large
Health Expenditure	21.89	<0.001	0.351	Large
Inflation	20.42	<0.001	0.335	Large

15.4 Country Lists by Cluster

Cluster 0: Intermediate Development Profile — 41 countries

#	Country	#	Country	#	Country
1	Albania	15	Georgia	29	Philippines
2	Armenia	16	Grenada	30	Samoa
3	Belize	17	Guatemala	31	Solomon Islands
4	Bhutan	18	Guyana	32	South Africa
5	Bolivia	19	Iraq	33	St. Vincent and the Grenadines
6	Botswana	20	Jamaica	34	Suriname
7	Cambodia	21	Jordan	35	Tonga
8	Cape Verde	22	Kyrgyz Republic	36	Tunisia
9	Congo, Rep.	23	Micronesia, Fed. Sts.	37	Turkmenistan
10	Dominican Republic	24	Moldova	38	Ukraine
11	Ecuador	25	Mongolia	39	Uzbekistan
12	El Salvador	26	Morocco	40	Vanuatu
13	Equatorial Guinea	27	Namibia	41	Vietnam
14	Fiji	28	Paraguay		

Cluster 1: High Development Profile — 31 countries

#	Country	#	Country	#	Country
1	Australia	12	France	23	Portugal
2	Austria	13	Germany	24	Serbia
3	Bahamas	14	Greece	25	South Korea
4	Barbados	15	Iceland	26	Spain
5	Bosnia and Herzegovina	16	Israel	27	Sweden
6	Canada	17	Italy	28	Switzerland
7	Chile	18	Japan	29	United Kingdom
8	Costa Rica	19	Netherlands	30	United States
9	Croatia	20	New Zealand	31	Uruguay
10	Denmark	21	Norway		
11	Finland	22	Poland		

Cluster 2: High Development Vulnerability — 44 countries

#	Country	#	Country	#	Country
1	Afghanistan	16	Guinea	31	Niger
2	Angola	17	Guinea-Bissau	32	Nigeria
3	Bangladesh	18	Haiti	33	Pakistan
4	Benin	19	India	34	Rwanda
5	Burkina Faso	20	Kenya	35	Senegal
6	Burundi	21	Kiribati	36	Sierra Leone
7	Cameroon	22	Lao	37	Sudan
8	Central African Republic	23	Lesotho	38	Tajikistan
9	Chad	24	Liberia	39	Tanzania
10	Comoros	25	Madagascar	40	Timor-Leste
11	Congo, Dem. Rep.	26	Malawi	41	Togo
12	Cote d'Ivoire	27	Mali	42	Uganda
13	Eritrea	28	Mauritania	43	Yemen
14	Gambia	29	Mozambique	44	Zambia
15	Ghana	30	Nepal		

Cluster 3: Trade-Intensive Development Profile — 27 countries

#	Country	#	Country	#	Country
1	Antigua and Barbuda	10	Ireland	19	Mauritius
2	Bahrain	11	Latvia	20	Montenegro
3	Belarus	12	Lebanon	21	Panama
4	Belgium	13	Lithuania	22	Seychelles
5	Bulgaria	14	Luxembourg	23	Singapore
6	Cyprus	15	Macedonia, FYR	24	Slovak Republic
7	Czech Republic	16	Malaysia	25	Slovenia
8	Estonia	17	Maldives	26	Thailand
9	Hungary	18	Malta	27	United Arab Emirates

Cluster 4: Higher-Inflation Development Profile — 24 countries

#	Country	#	Country	#	Country
1	Algeria	9	Gabon	17	Peru
2	Argentina	10	Indonesia	18	Qatar
3	Azerbaijan	11	Iran	19	Romania
4	Brazil	12	Kazakhstan	20	Russia
5	Brunei	13	Kuwait	21	Saudi Arabia
6	China	14	Libya	22	Sri Lanka
7	Colombia	15	Myanmar	23	Turkey
8	Egypt	16	Oman	24	Venezuela

Summary

Cluster	Profile	Countries
0	Intermediate Development	41
1	High Development	31
2	High Development Vulnerability	44
3	Trade-Intensive Development	27
4	Higher-Inflation Development	24
Total		167

This comprehensive report combines exploratory data analysis with rigorous clustering methodology to provide actionable insights for development policy and resource allocation.